Vizualizacija postupaka učenja neuronskih mreža

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Sadržaj

- Neuronske mreže
- Vizualiziranje
- Postojeće tehnike
- Implementacija
- Primjena
- Zaključak

Neuronske mreže

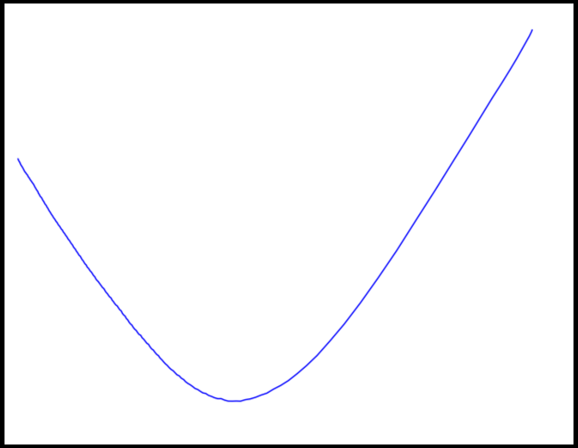
- veliki broj varijabli (težinski faktori)
- kompleksan sustav
- teško razumjevanje promjene kroz učenje

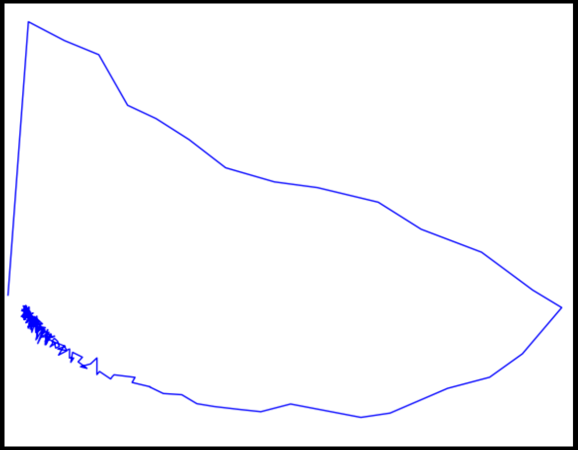
Vizualiziranje

- moćan alat za razumjevanje kompleksnih sustava
- teorija kaosa
- veliki broj dimenzija neuronske mreže
- veliki broj vizualizacija

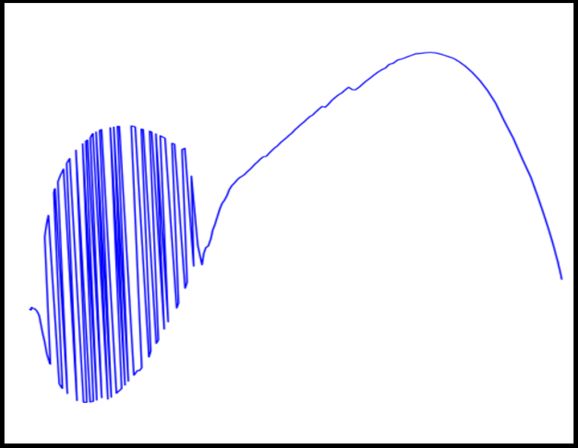
Postojeće tehnike 1

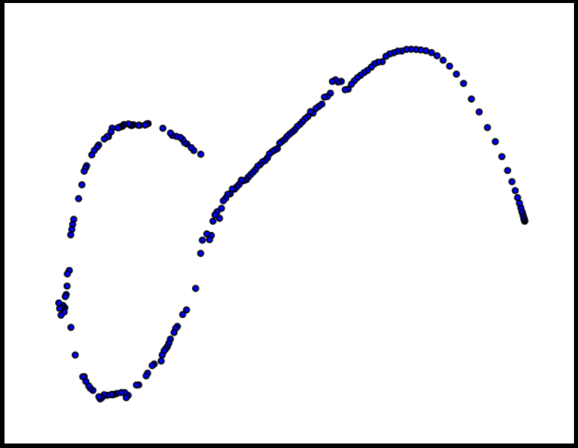
PCA





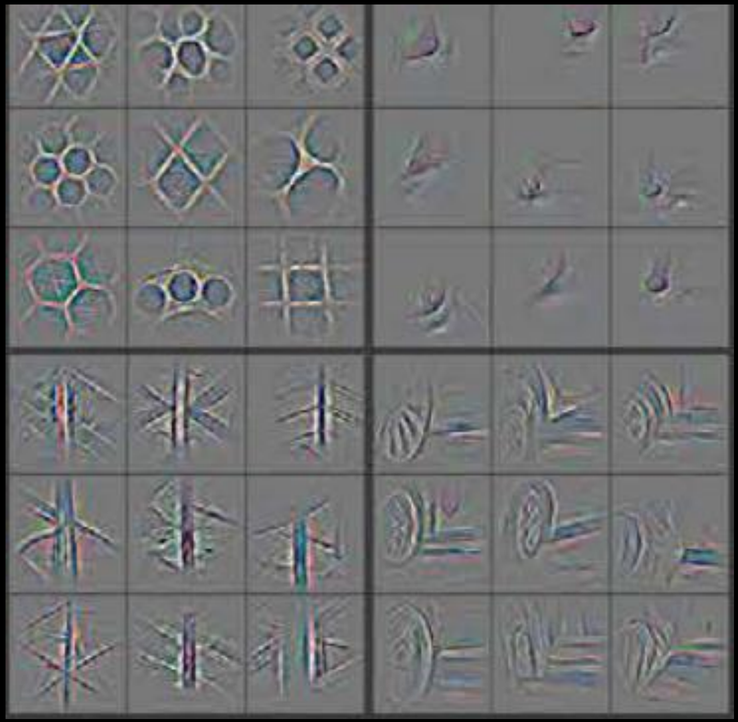
t-SNE





Postojeće tehnike 2

- vizualiziranje aktivacija konvolucije
- generiranje slike



Implementacija

- Python, tensorflow
- testiranje se radilo sa:
    - neuronska mreža za učenje word2vec reprezentacije riječi
    - neuronska mreža za raspoznavanje pisanih brojeva
    - neuronska mreža koja je naučila logičko pravilo XOR

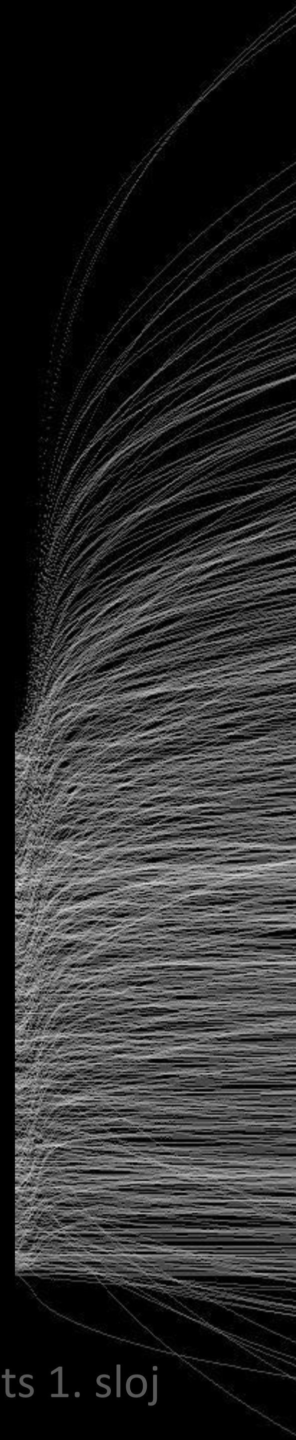
Implementirane tehnike

- absolute
- norm_by_all
- norm_by_row
- norm_by_col
- diff
- adding
- sort
- rank
- average
- vector_magnitude
- vector_dist_deviation
- vector_angle_diff
- vector_angle_deviation
- angle_path
- plot
- plot2

Primjena

```
import neurovis as nv
vis = nv.Vis(”log”)
vis.load( nv.Basic( l=1, n=[-1], inout=”in” ), [-1] )
vis.backup()
```

```
vis.restore()
vis.plot()
vis.norm_by_all(log=True)
vis.save()
```

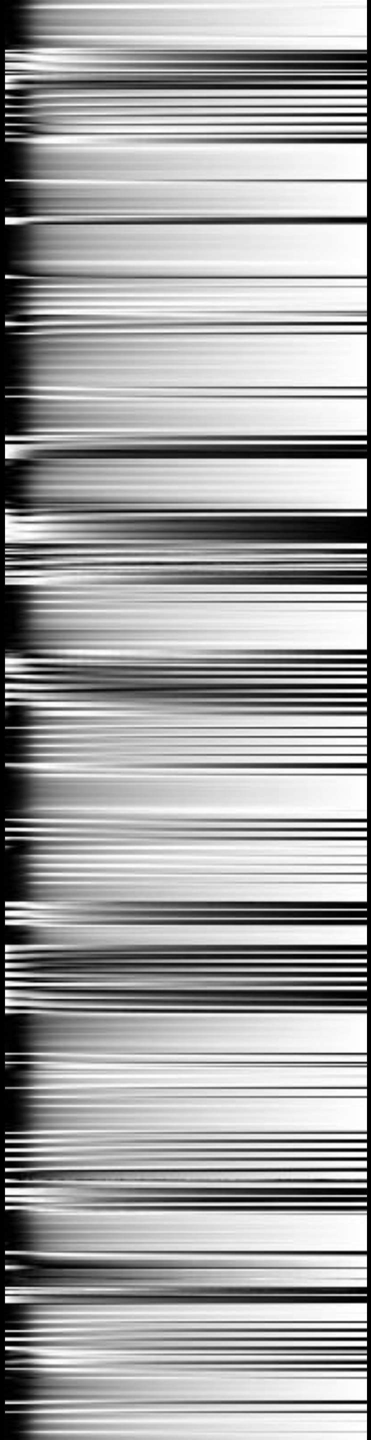


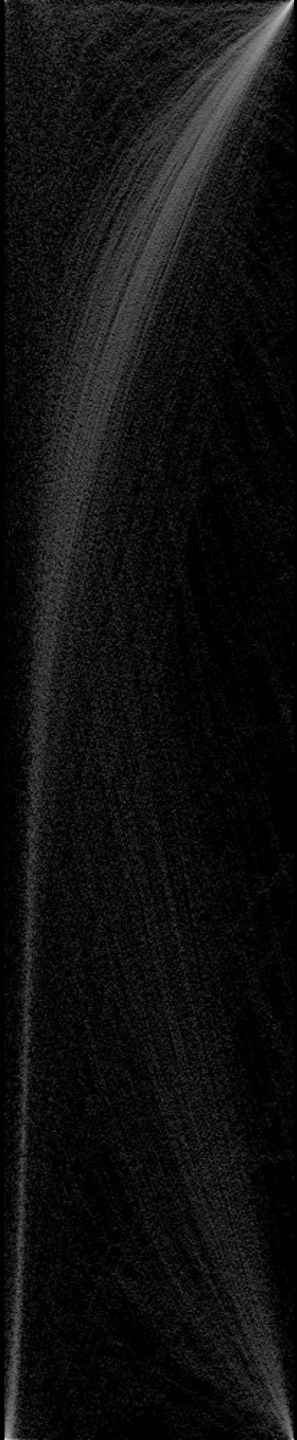
Digits 1. sloj

Word2vec 50. neuron

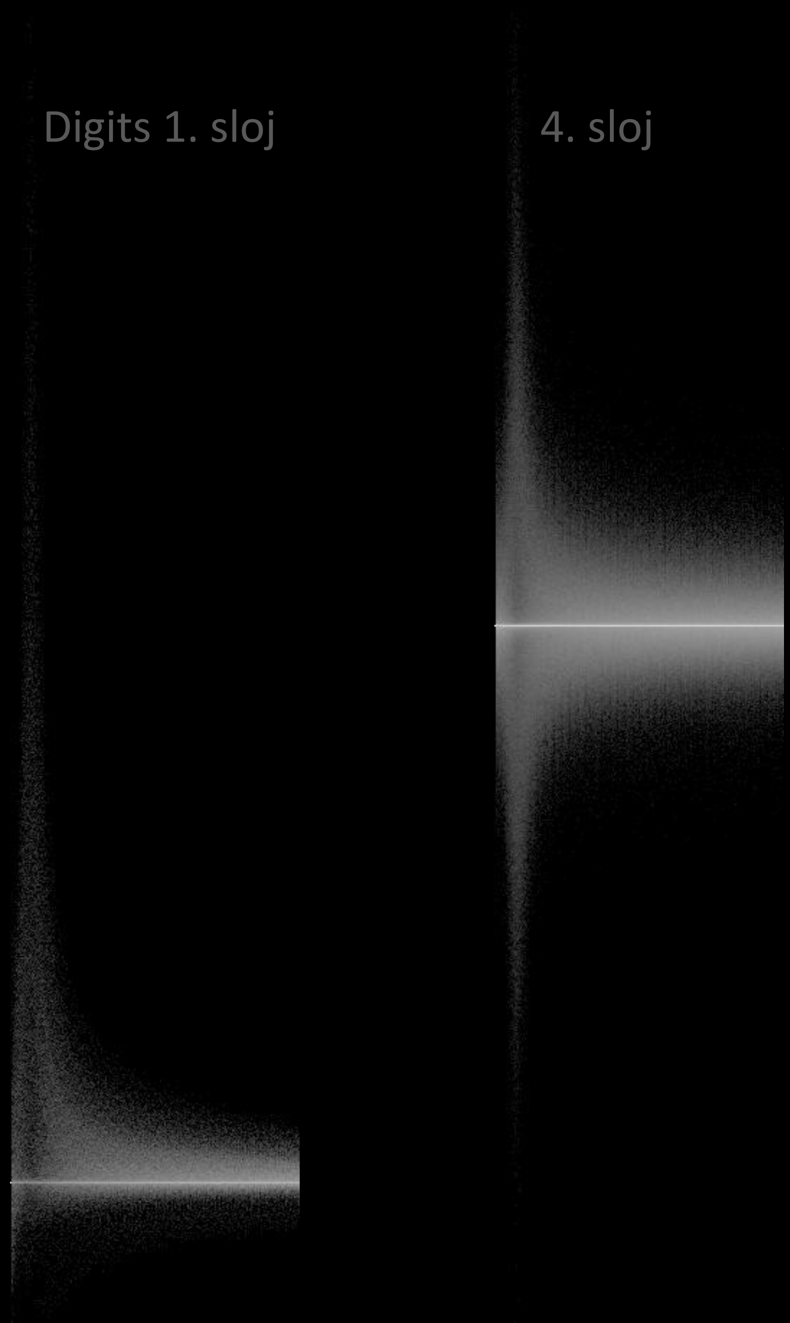
Digits 1. sloj

```
vis.restore()
vis.norm_by_row()
vis.plot()
vis.norm_by_all(log=True)
vis.save()
```

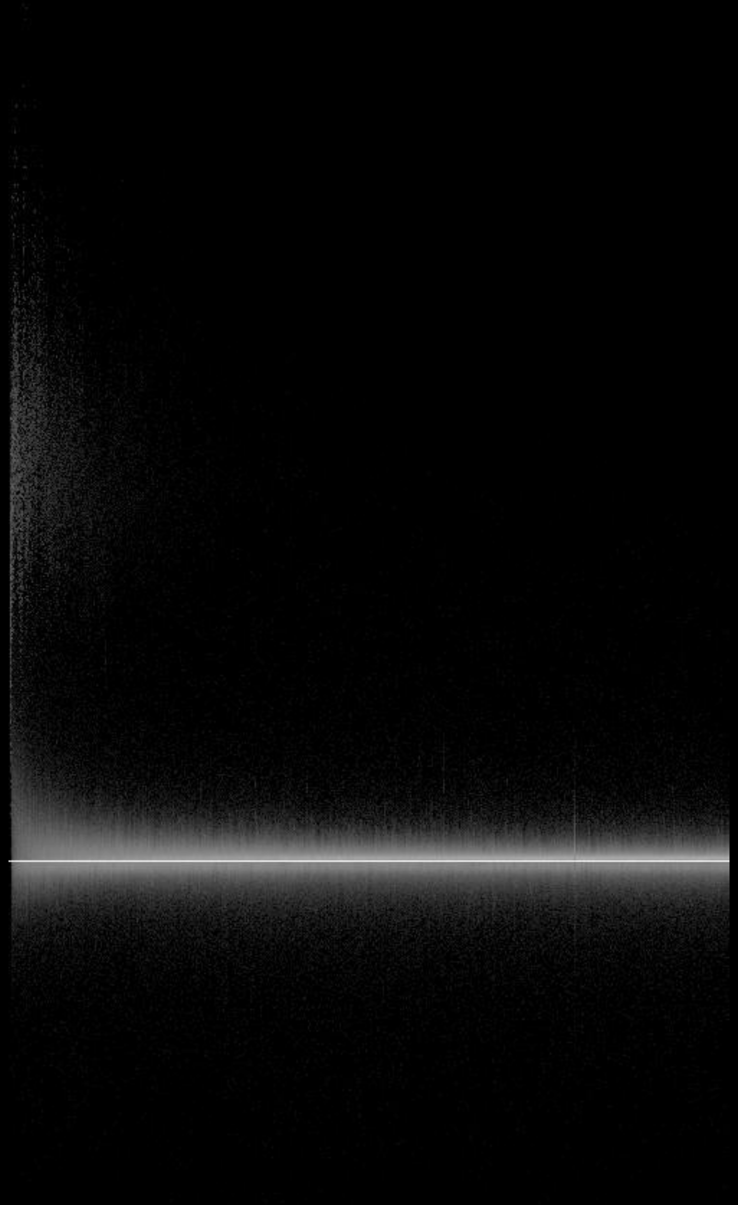




```
vis.restore()
vis.diff(1)
vis.plot()
vis.norm_by_all(log=True)
vis.save()
```



```
vis.restore()
vis.diff(1)
vis.plot()
vis.norm_by_all(log=True)
vis.save()
```

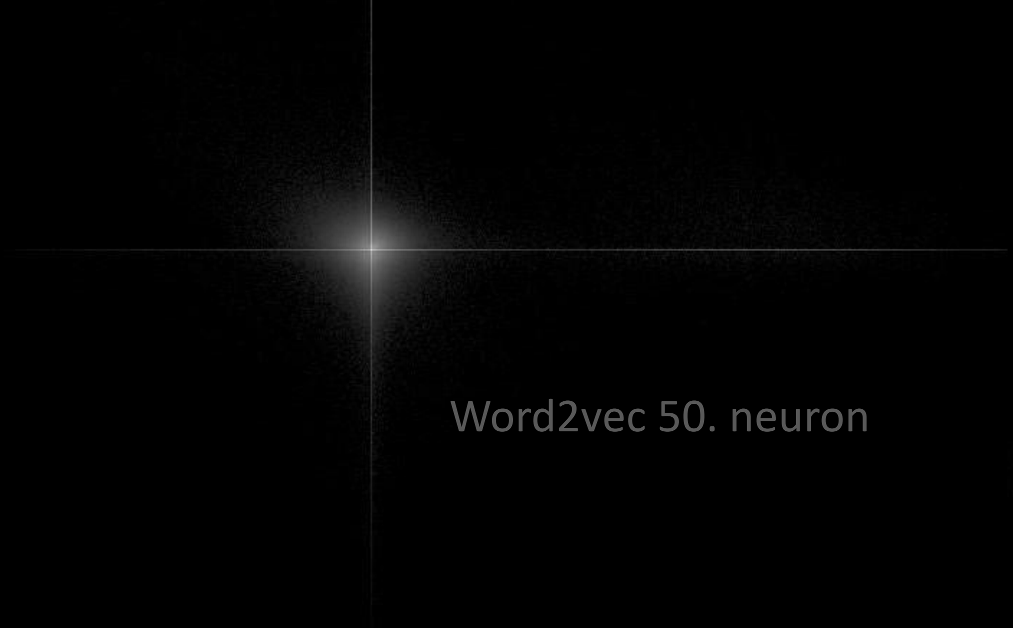


Word2vec 50. neuron

```
vis.restore()
vis.diff(1)
vis.plot2()
vis.norm_by_all(log=True)
vis.save()
```

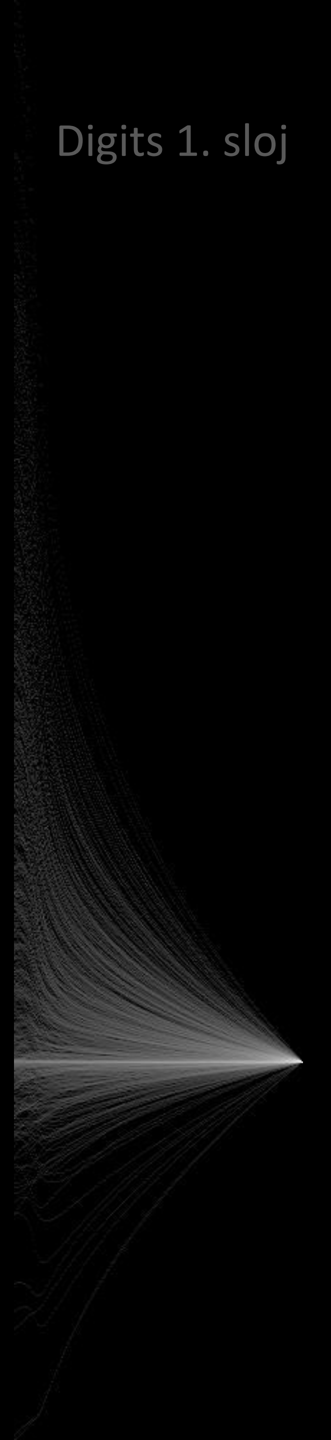
Digits 1. sloj

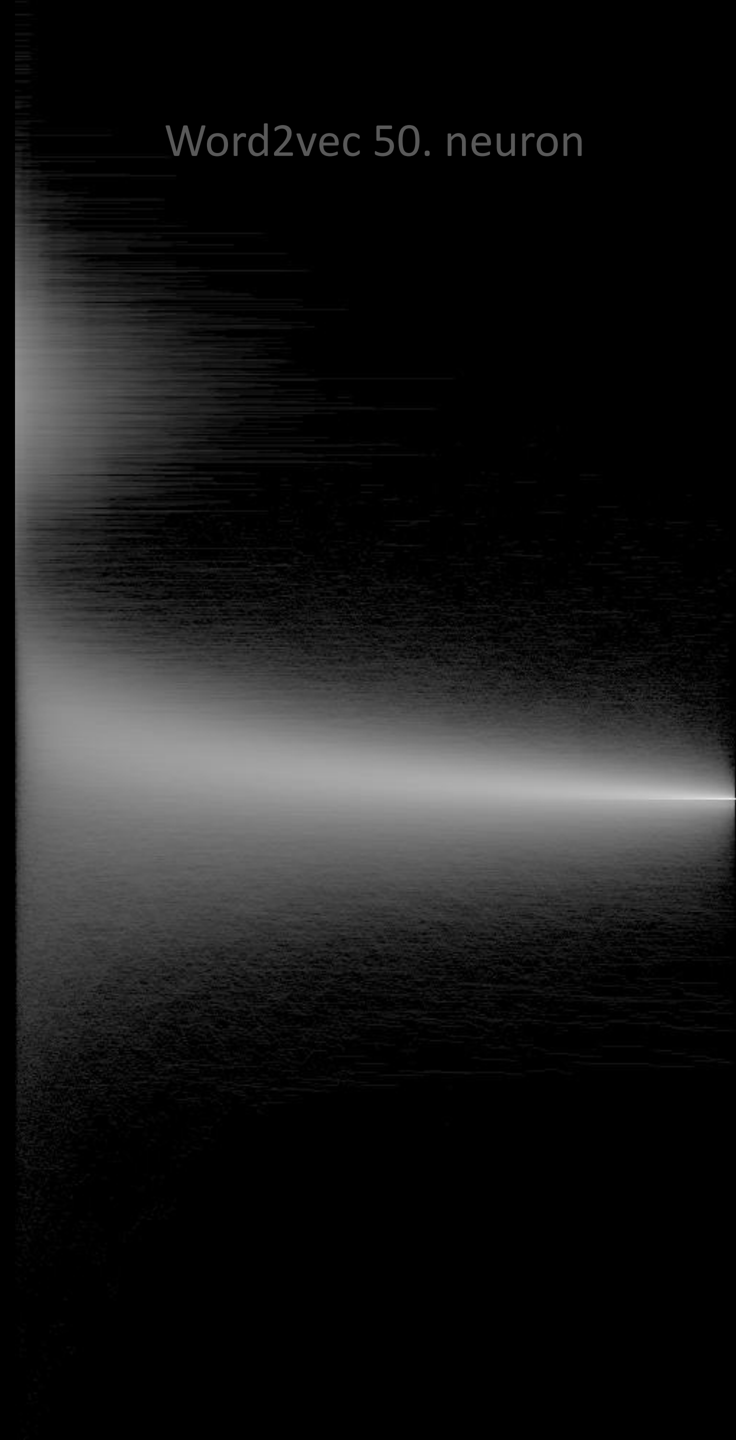
Digits 4. sloj



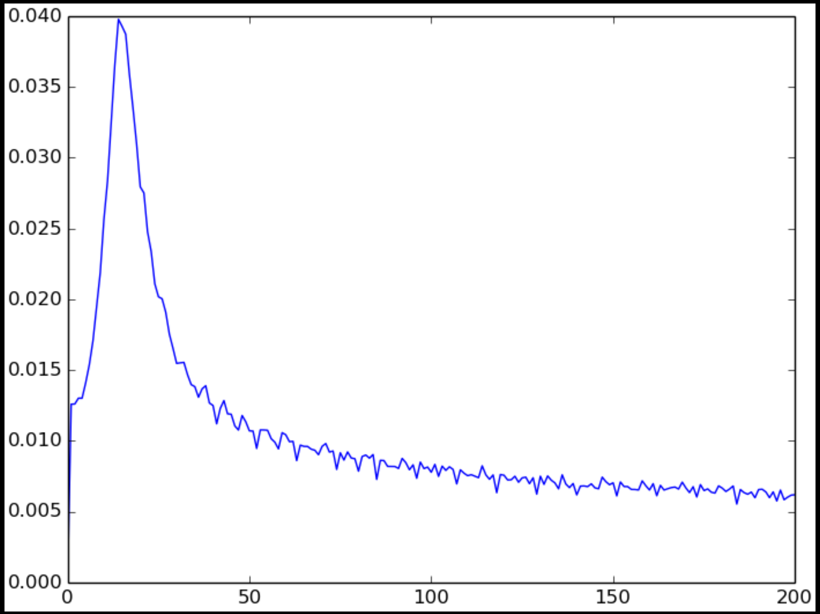
Word2vec 50. neuron

```
vis.restore()
vis.diff(-1)
vis.plot()
vis.norm_by_all(log=True)
vis.save()
```

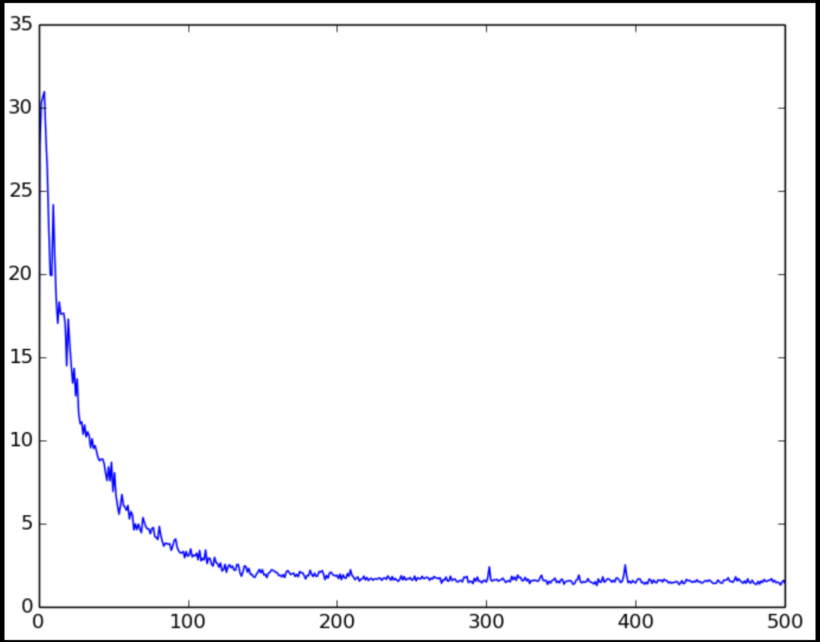




```
vis.restore()
vis.diff(1)
vis.vector_magnitude()
vis.plot_show()
```

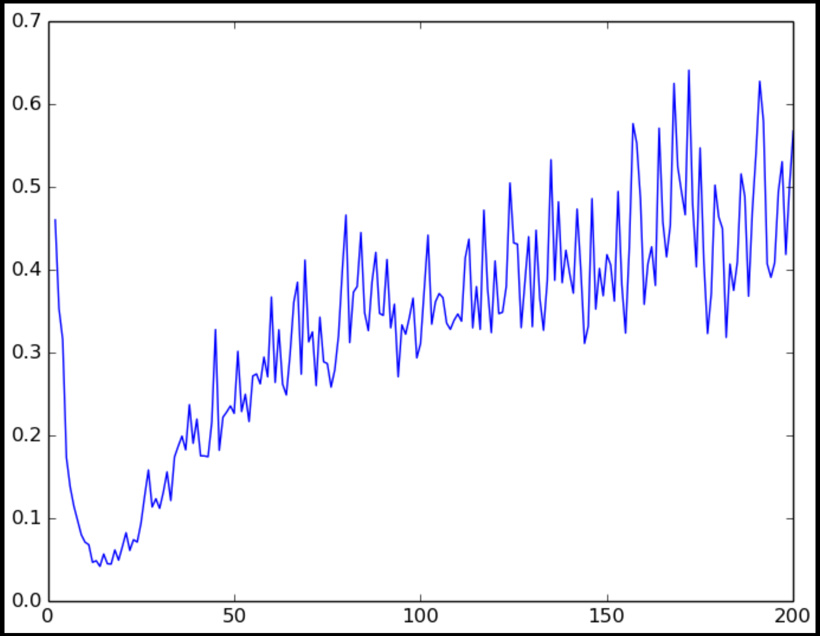


Digits 4. sloj

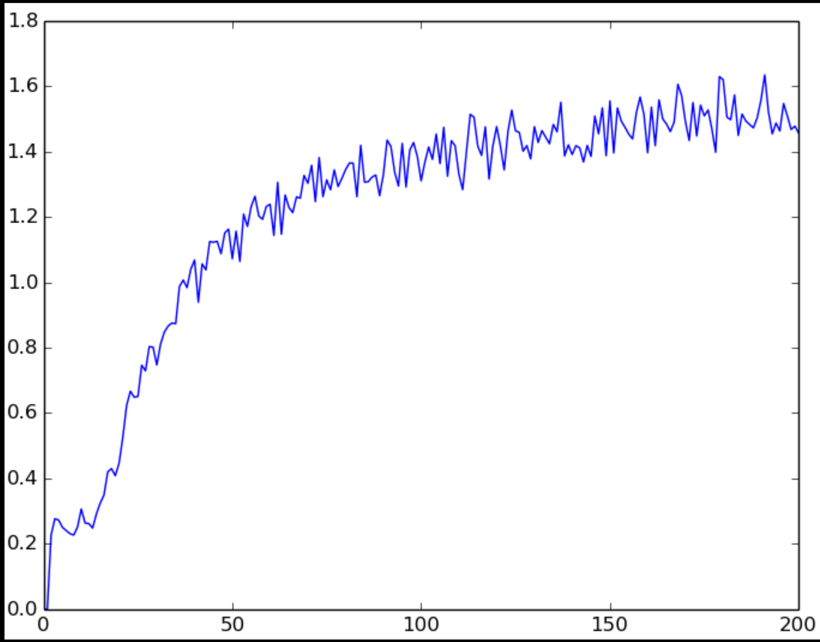


Word2vec 50. neuron

```
vis.restore()
vis.diff(1)
vis.vector_angle_diff()
vis.plot_show()
```

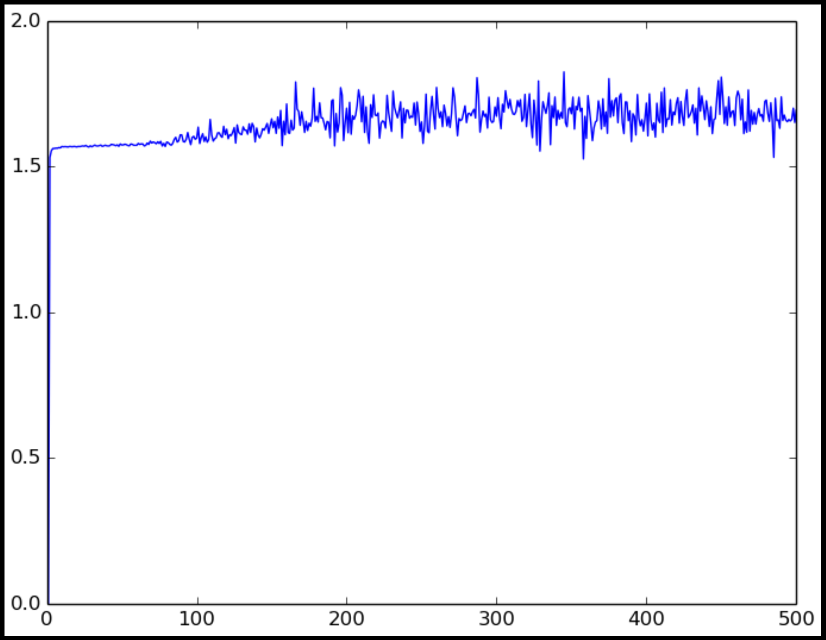


Digits 1. sloj

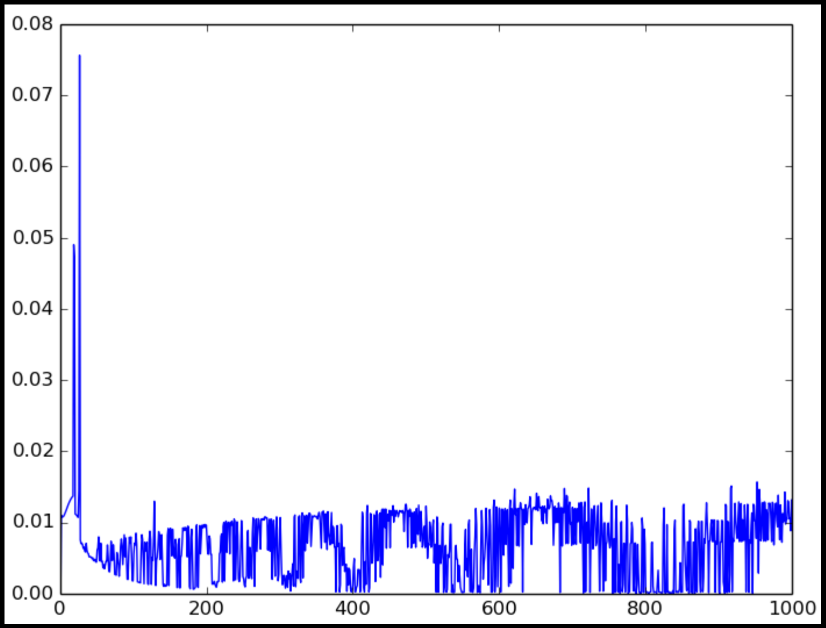


Digits 4. sloj

```
vis.restore()
vis.diff(1)
vis.vector_angle_diff()
vis.plot_show()
```

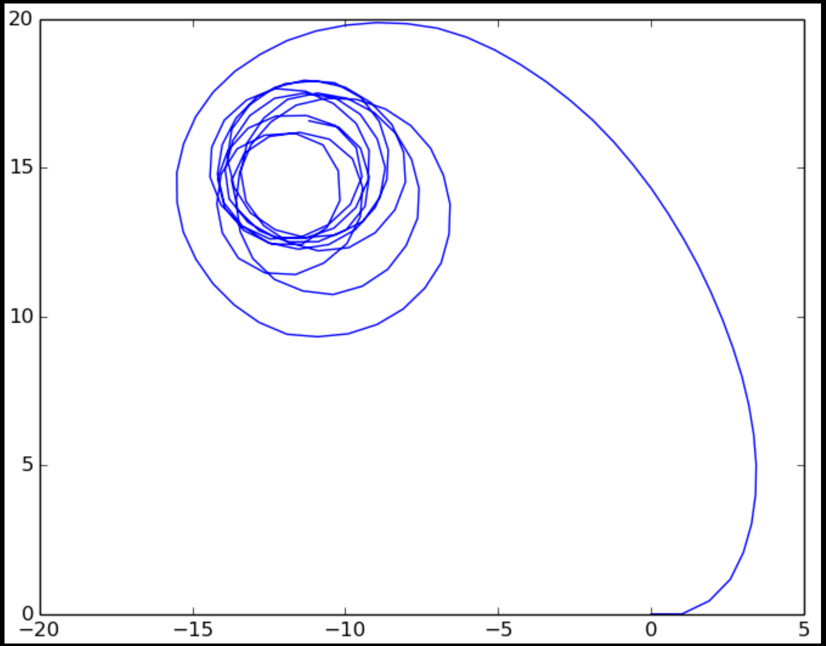


Word2vec 50. neuron

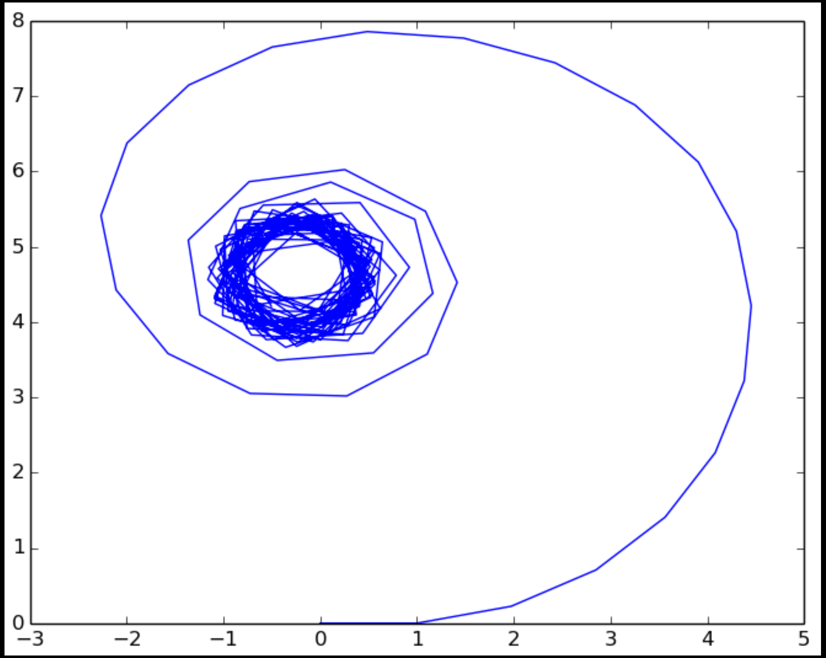


XOR

```
vis.restore()
vis.diff(1)
vis.vector_angle_diff()
vis.angle_path()
vis.plot_show(xy=True)
```

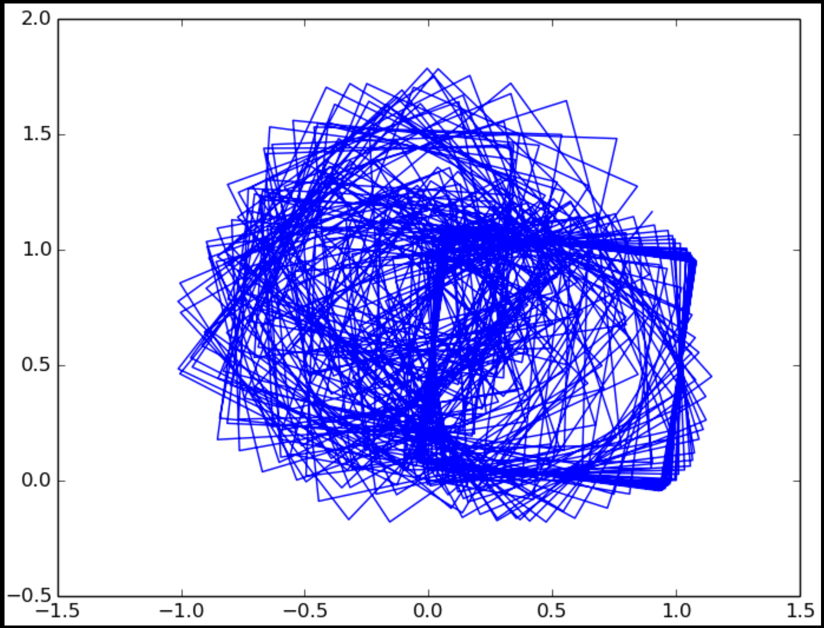


Digits 1. sloj

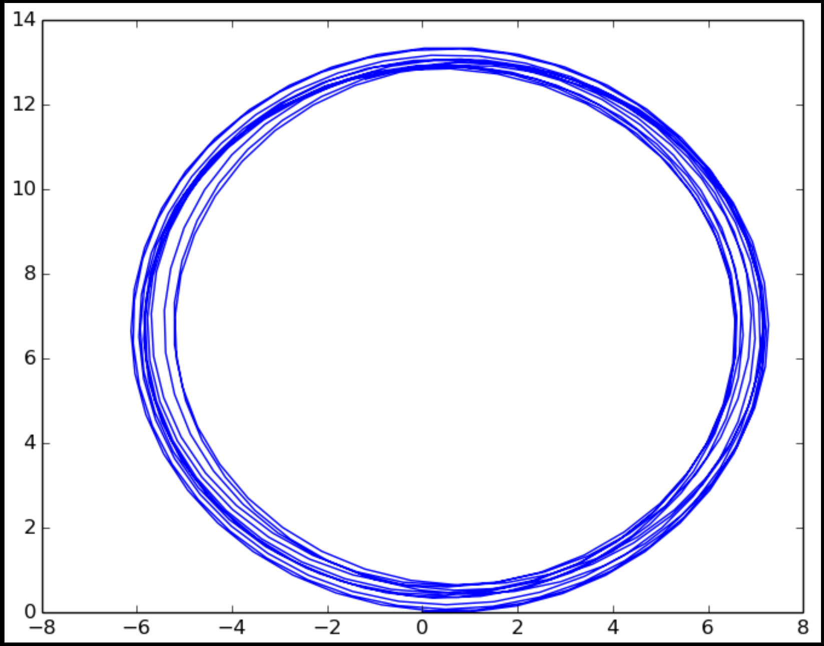


Digits 4. sloj

```
vis.restore()
vis.diff(1)
vis.vector_angle_diff()
vis.angle_path(path=10)
vis.plot_show(xy=True)
```

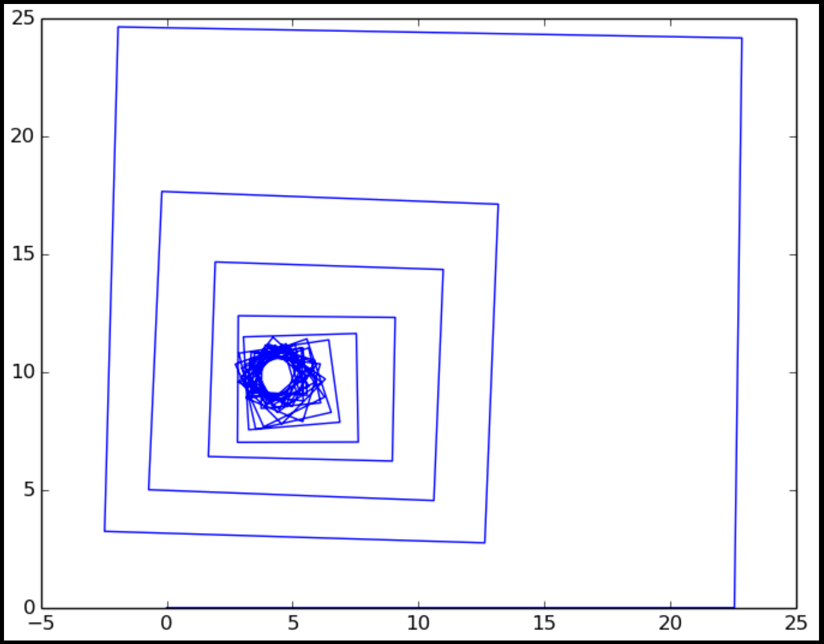


Word2vec 50. neuron

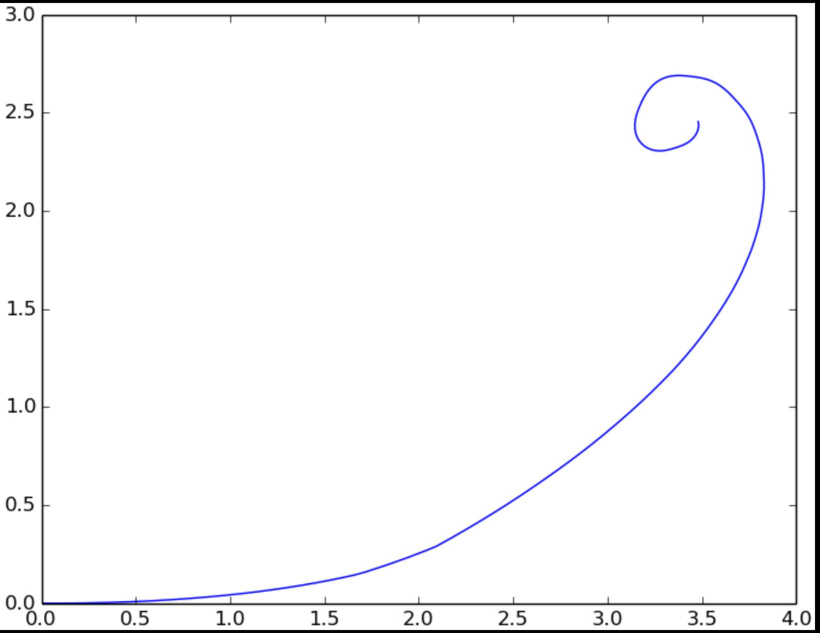


path=10

```
vis.restore()
vis.diff(1)
vis.vector_magnitude()
magnitude = vis.v.copy()
vis.restore()
vis.diff(1)
vis.vector_angle_diff()
vis.angle_path(
  step=magnitude)
vis.plot_show(xy=True)
```

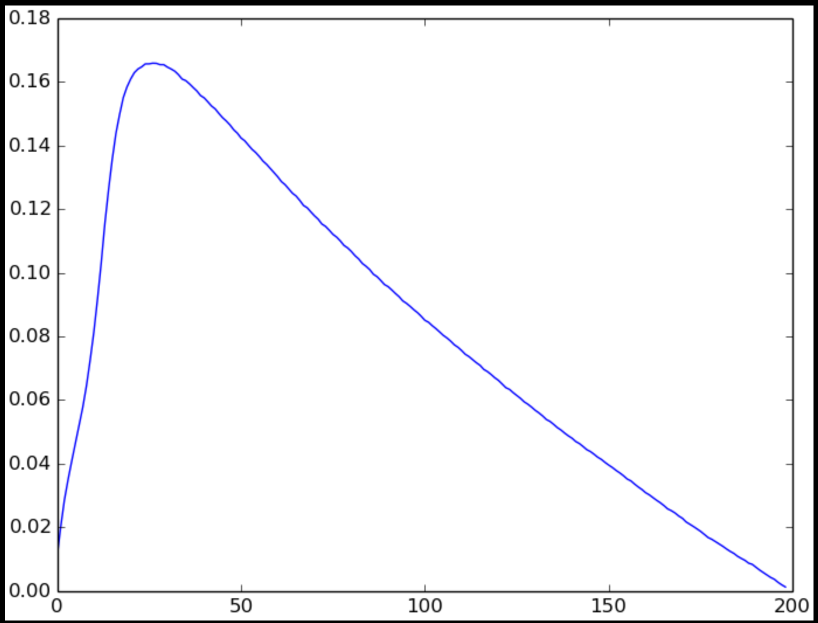


Word2vec 50. neuron

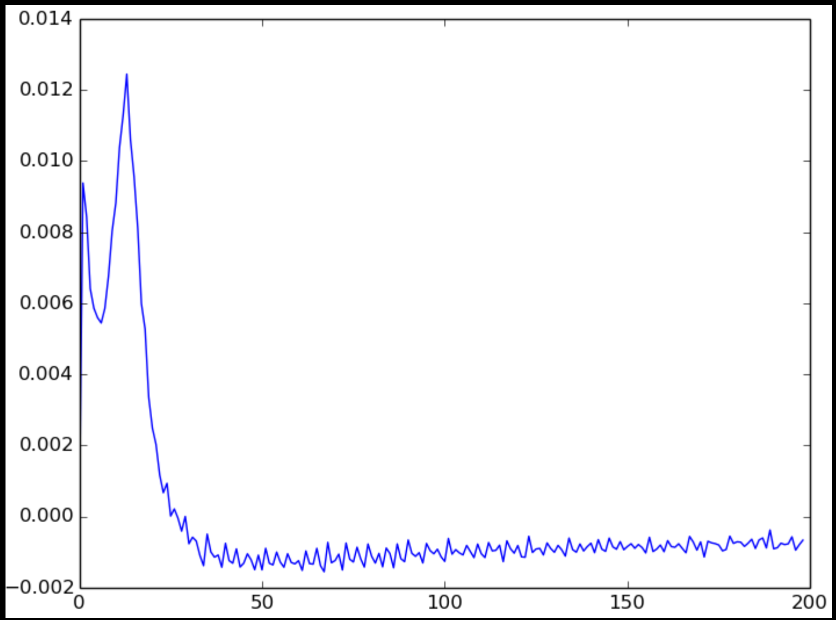


XOR

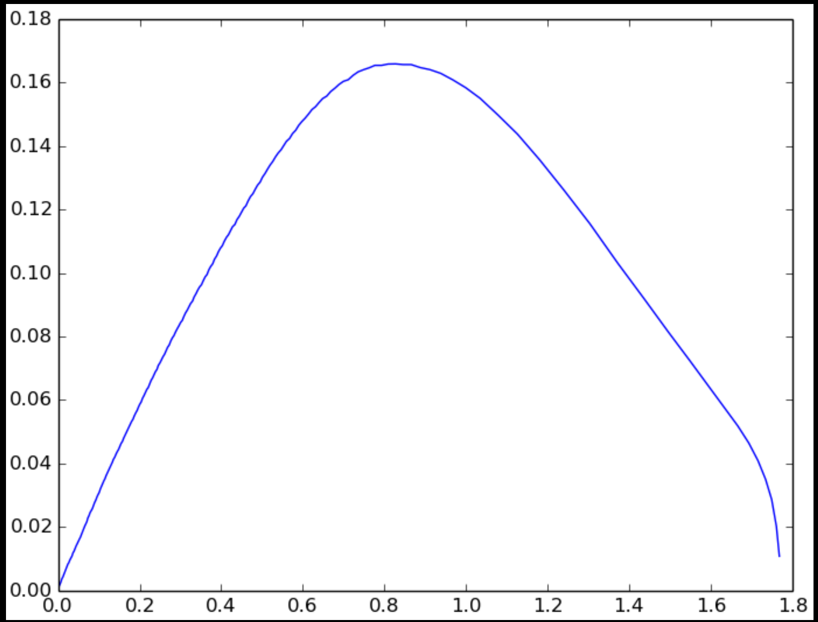
```
vis.restore()
vis.vector_dist
  _deviation()
vis.plot_show()
```



Digits 1. sloj

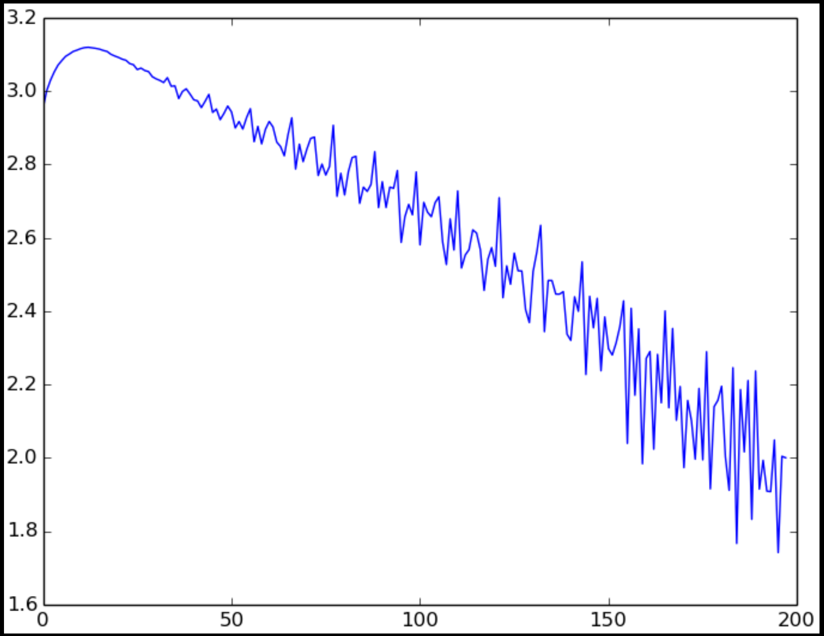


diff(1)

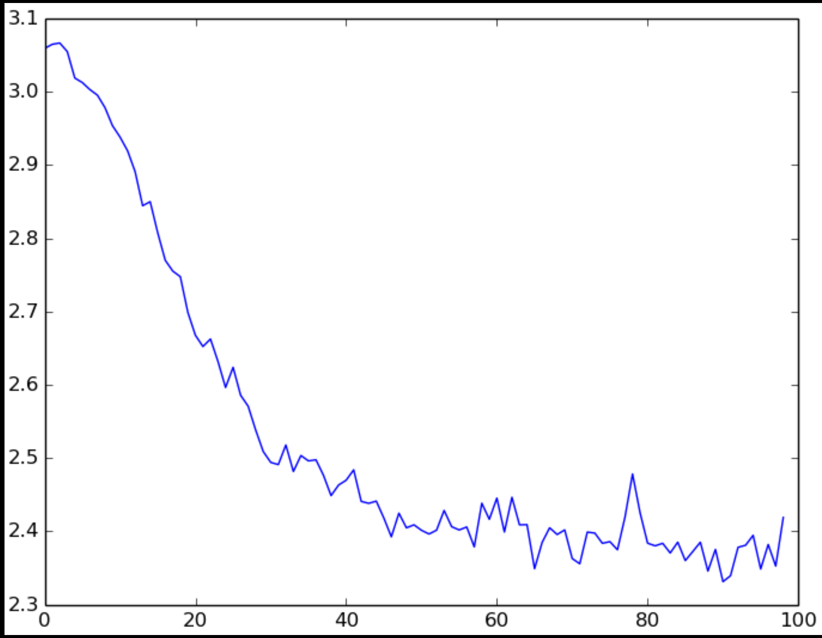


y="vec-dist-devi", x="diff(-1),vec-mag"

```
vis.restore()
vis.diff(1)
vis.vector_angle
  _deviation()
vis.plot_show()
```

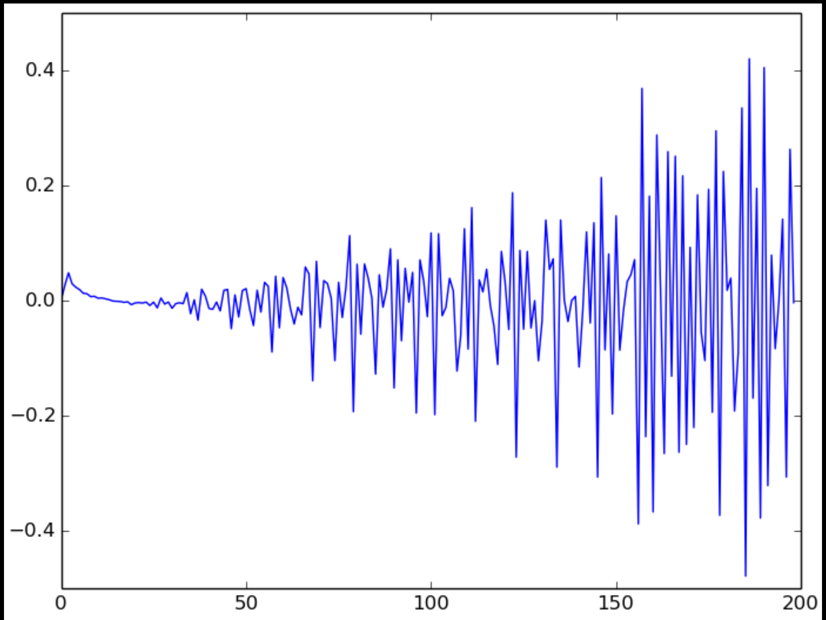


Digits 1. sloj

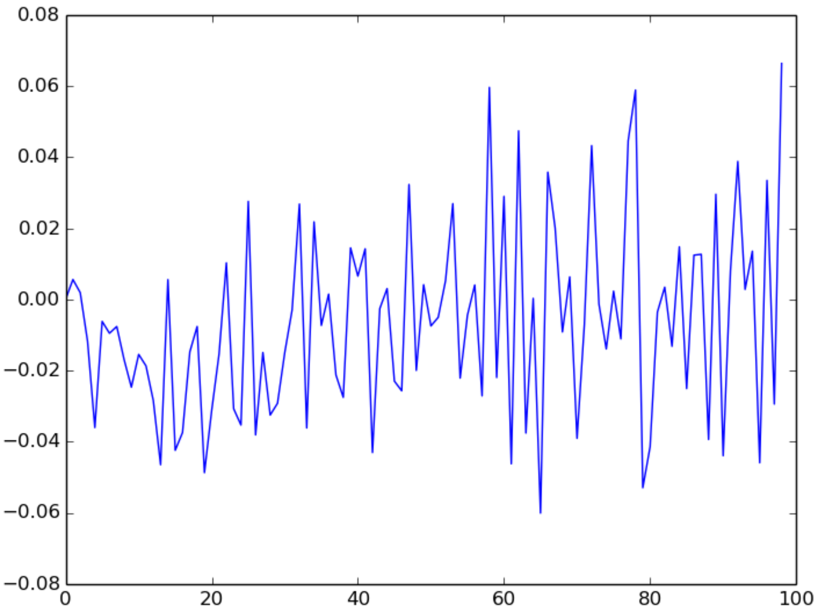


Word2vec 50. neuron

```
vis.restore()
vis.diff(1)
vis.vector_angle
  _deviation()
vis.diff(1)
vis.plot_show()
```



Digits 1. sloj



Word2vec 50. neuron

Zaključak

- Kombiniranje transformacija rezultiralo velikim brojem vizualizacija
- Izrada novih vizualizacija:
  - određenim kombinacijama koje korisnik zna da želi upotrijebiti
  - otkrivanjem vizualizacija nasumičnim kombiniranjem
- Mogućnost kompleksnijih tehnika transformacija:
  - ako se omogući procesiranje velike količine podataka